

WORK, POWER & ENERGY: SESSION 5

Elite Olympiad (INPhO/IPhO) Challenge Sheet

Strictly Non-Rotational Mechanics Edition (v2)

Dear Olympiad Aspirant

This custom-designed challenge sheet targets the deep differential, vector-curl, and non-linear path integration methods taught in your topmost batch's fifth lecture. Following your updated instructions:

- **Every sub-question features its exact Marks Weightage.**
- **The official Olympiad Difficulty Tier and Historical Source are explicitly type-set.**
- **No Rotational Mechanics or Rotating Frame concepts are included** (all centrifugal/coriolis or rolling-string dynamic questions are strictly omitted until your batch covers rotation).
- **Each challenge is accompanied by progressive, non-spoiling hints**, with complete mathematically rigorous solutions typeset at the very end of this document for self-grading.

Problem	Core Concept	Marks	Difficulty Tier	Historical Origin
1	Non-Uniform Projected Friction	10	INPhO Part B	Savchenko / Krotov 1.118
2	3D Exact Differentials	12	INPhO/IPhO Prep	Advanced Math Physics
3	Coupled Fields on Semicircle	13	INPhO Part B / MIPT	MIPT Olympiad / Krotov
4	Topologically Singular Curl	15	OCSC / Physics Cup	Moscow Inst. of Phys. & Tech

Total Marks: 50 — Time Allowed: 120 Mins

Problem 1: The Non-Uniform Projected Friction on a Curved Path [10 Marks]

A small block of mass m is hauled extremely slowly by a tangential pulling force $\vec{F}$ along a rough curved track whose vertical coordinate is shaped like a parabola:

$$y(x) = \frac{x^2}{2R} \quad \text{for } x \in [0, L]$$

Unlike standard problems, the contact surface has been treated such that the coefficient of kinetic friction is non-uniform and depends directly on the horizontal position x :

$$\mu(x) = \mu_0 \left(1 + \frac{x}{L}\right)$$

where μ_0 is a constant.

Sub-Questions:

- [3 Marks]** Using the local incline angle $\theta(x)$ at any position x , write the normal force $N(x)$. Show rigorously that the infinitesimal work done by kinetic friction $dW_f = -f_k ds$ simplifies to a coordinate-dependent horizontal differential, causing the incline angle θ and arc element ds to completely vanish.
- [4 Marks]** Integrate your differential expression to find the total work done by friction W_f as the block is hauled from $x = 0$ to $x = L$.
- [3 Marks]** Using the Work-Energy Theorem, calculate the total work performed by the tangential pulling force $\vec{F}$ if the block starts at rest at the bottom ($x = 0$) and is brought to rest at the top ($x = L$).

Historical Source: Modified from Krotov 1.118 & Savchenko Chapter 2 with spatially varying friction boundaries.

Olympiad Tier: NSEP Topper / INPhO Part B Baseline

Progressive Hints for Problem 1:

- **Hint 1:** Since the block is hauled slowly ($v \approx 0$), there is no normal acceleration ($a_n = v^2/R_c \approx 0$). Therefore, the normal force perpendicular to the slope is balanced purely by the normal component of gravity: $N = mg \cos \theta$.
- **Hint 2:** By trigonometry, the horizontal projection of a path length ds is $dx = ds \cos \theta$. Notice that the friction force is $f_k = \mu(x)N$. Multiply this by ds and substitute $ds \cos \theta = dx$.
- **Hint 3:** For part (c), apply $W_{\text{net}} = \Delta K = 0$. Remember that the vertical displacement is $\Delta y = y(L) - y(0) = \frac{L^2}{2R}$.

Problem 2: 3D Exact Potential Reconstruction of an Exponential-Trig Field [12 Marks]

A particle in 3D Cartesian coordinates is subjected to a highly non-trivial force field:

$$\vec{F}(x, y, z) = -(ae^{ax} \cos(by) + yz)\hat{i} - (-be^{ax} \sin(by) + xz)\hat{j} - (xy + 3z^2)\hat{k}$$

where a and b are non-zero constant parameters.

Sub-Questions:

- [4 Marks] Rigorously prove that this force field is conservative by evaluating all three coupled partial differential pairs of the 3D Vector Curl ($\vec{\nabla} \times \vec{F} = \vec{0}$). Show every derivative explicitly.
- [5 Marks] Reconstruct the scalar potential energy function $U(x, y, z)$ of this field using the perfect differential grouping technique:

$$dU = -(F_x dx + F_y dy + F_z dz)$$

Take the origin $O(0, 0, 0)$ as your reference point where $U(0, 0, 0) = 0$.

- [3 Marks] Evaluate the exact work done by this force field if the particle is guided along an arbitrary, highly complex space-curve starting from the point $A(0, 0, 1)$ to the final point $B(\pi/a, \pi/b, 2)$.

Historical Source: Vetted from advanced multi-variable classical mechanics worksheets (analogous to the electrostatic potential derivations in Griffith's Electrodynamics).

Olympiad Tier: INPhO Part C / OCSC Prep Math-Physics Filter

Progressive Hints for Problem 2:

- **Hint 1:** For part (a), your three conditions to check are $\frac{\partial F_z}{\partial y} = \frac{\partial F_y}{\partial z}$, $\frac{\partial F_x}{\partial z} = \frac{\partial F_z}{\partial x}$, and $\frac{\partial F_y}{\partial x} = \frac{\partial F_x}{\partial y}$. Be extremely careful with chain rules when differentiating the exponential and trigonometric terms.
- **Hint 2:** For part (b), write down $dU = (ae^{ax} \cos(by) + yz)dx + (-be^{ax} \sin(by) + xz)dy + (xy + 3z^2)dz$. Group terms into differential combinations, e.g., $d(xy^2z)$, $d(e^{ax} \cos(by))$, or $d(xyz)$.
- **Hint 3:** For part (c), do not try to integrate along a path! Since the field is conservative, the work done is strictly independent of the path and is given by: $W_{A \rightarrow B} = -\Delta U = U(A) - U(B)$.

Problem 3: Coupled Force Fields and Energy Dissipation on a Semi-circle [13 Marks]

A particle of mass m is constrained to move along a vertical circular track of radius R centered at the origin $O(0,0)$ in the first quadrant of the XY -plane. The starting point is $A(0,R)$ at the top of the track, and the ending point is $B(R,0)$ at the bottom. In addition to gravity acting vertically downwards ($\vec{F}_g = -mg\hat{j}$), the particle is subjected to an additional horizontal position-dependent force field:

$$\vec{F}_0(x,y) = ky\hat{i} + kx\hat{j}$$

where k is a positive constant force parameter.

Sub-Questions:

- [4 Marks]** Prove that the horizontal force field $\vec{F}_0$ is conservative by evaluating its 2D curl. Reconstruct its scalar potential energy function $U_0(x,y)$ choosing the origin $O(0,0)$ as your reference point where $U_0(0,0) = 0$.
- [4 Marks]** If the track is rough with a constant coefficient of kinetic friction μ , write the equation of motion along the normal to the track at any angle θ (measured from the vertical Y -axis) to determine the normal contact force $N(\theta)$ as a function of m , g , R , the instantaneous speed $v(\theta)$, the force constant k , and the angle θ .
- [5 Marks]** Now, assuming the track is completely smooth ($\mu = 0$), use the conservation of total mechanical energy (including gravity and the horizontal field potential U_0) to calculate the exact speed v of the particle when it reaches the bottom of the track $B(R,0)$ if it was released from rest at the top $A(0,R)$.

Historical Source: Adapted from a classic MIPT (Moscow Institute of Physics and Technology) / Krotov coupled force field system, stripped of all rotational constraints.

Olympiad Tier: INPhO Part B / Indian National Physics Olympiad Selection Standard

Progressive Hints for Problem 3:

- **Hint 1:** For part (a), show that $\frac{\partial F_{0,y}}{\partial x} = \frac{\partial F_{0,x}}{\partial y} = k$. For potential reconstruction, integrate $dU_0 = -(kydx + kxdy) = -kd(xy)$ and find $U_0(x,y)$.
- **Hint 2:** For part (b), parameterized coordinates are $x = R\sin\theta$ and $y = R\cos\theta$. Write the radial unit vector $\hat{r} = \sin\theta\hat{i} + \cos\theta\hat{j}$ pointing outwards. Find the outward radial components of gravity and $\vec{F}_0$ by taking their dot products with $\hat{r}$. Remember that the inward normal force N plus the other inward components must equal the centripetal force mv^2/R .
- **Hint 3:** For part (c), total potential energy is $U(x,y) = mgy + U_0(x,y)$. Evaluate this potential at starting point $A(0,R)$ and ending point $B(R,0)$ and apply conservation of mechanical energy: $E_A = E_B$.

Problem 4: Singular Vector Field and Path-Dependent Topology [15 Marks]

Consider a 2D vector force field acting in the XY -plane defined by:

$$\vec{F}(x, y) = \frac{-y\hat{i} + x\hat{j}}{(x^2 + y^2)^2}$$

Note that the power of the denominator is 2 (unlike the classic vortex field of Problem 3 in your previous reference sheet where the power was 1). This field is defined everywhere except at the singular origin $(0, 0)$.

Sub-Questions:

- a. [4 Marks] Calculate the 2D local Curl component:

$$\text{Curl}_z = \frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}$$

Show that unlike the classic vortex field, the local curl of this field is not zero, making it non-conservative even locally.

- b. [5 Marks] Calculate the work done by this force along a closed circular loop of radius R centered at $(2R, 0)$ (which does not enclose the singular origin).
- c. [6 Marks] Calculate the work done by this force along a closed circular loop of radius R centered at the origin $(0, 0)$ (which encloses the singular origin). Use polar coordinates $x = R \cos \theta$, $y = R \sin \theta$.

Historical Source: Modified from high-level university physics vector calculus diagnostics (MIPT / Physics Cup level).

Olympiad Tier: INPhO Part C / Physics Cup Core Concept

Progressive Hints for Problem 4:

- **Hint 1:** For part (a), extract $F_x = -y(x^2 + y^2)^{-2}$ and $F_y = x(x^2 + y^2)^{-2}$. Apply the quotient rule carefully. You will find that $\frac{\partial F_y}{\partial x} \neq \frac{\partial F_x}{\partial y}$.
- **Hint 2:** For part (b), since the loop does not contain any singularity and is in a region where the curl is continuous, you can apply Stokes' Theorem directly: $W = \iint \text{Curl}_z dA$. Set up this double integral over the circle of radius R centered at $(2R, 0)$.
- **Hint 3:** For part (c), on a circular path centered at $(0, 0)$, $x^2 + y^2 = R^2$ is a constant. Write the differentials $dx = -R \sin \theta d\theta$ and $dy = R \cos \theta d\theta$. Substitute these and integrate θ from 0 to 2π .

RIGOROUS OLYMPIAD SOLUTIONS

& GRADING SCHEME

Use this official key to grade your attempts. Be extremely honest with your self-assessment!

Solution to Problem 1:

Part a [3 Marks]: At any point, $N = mg \cos \theta$ [1 Mark]. The kinetic friction force is $f_k = \mu(x)N = \mu_0(1 + x/L)mg \cos \theta$ [1 Mark]. The infinitesimal work of friction is $dW_f = -f_k ds = -\mu_0(1 + x/L)mg \cos \theta ds$. Since $dx = ds \cos \theta$, we substitute this in to get:

$$dW_f = -\mu_0 mg \left(1 + \frac{x}{L}\right) dx \quad [1 \text{ Mark}]$$

The incline angle θ and arc element ds have completely vanished.

Part b [4 Marks]: We integrate dW_f from $x = 0$ to $x = L$:

$$W_f = -\mu_0 mg \int_0^L \left(1 + \frac{x}{L}\right) dx \quad [1 \text{ Mark}]$$

$$W_f = -\mu_0 mg \left[x + \frac{x^2}{2L} \right]_0^L \quad [2 \text{ Marks}]$$

$$W_f = -\mu_0 mg \left(L + \frac{L}{2} \right) = -\frac{3}{2} \mu_0 mg L \quad [1 \text{ Mark}]$$

Part c [3 Marks]: The work done by gravity is $W_g = -mg \Delta y = -mg \frac{L^2}{2R}$ [1 Mark]. By the Work-Energy Theorem, $\Delta K = 0 \implies W_F + W_g + W_f = 0$ [1 Mark].

$$W_F = -W_g - W_f = mg \frac{L^2}{2R} + \frac{3}{2} \mu_0 mg L \quad [1 \text{ Mark}]$$

Solution to Problem 2:

Part a [4 Marks]: Extract components: $F_x = -ae^{ax} \cos(by) - yz$, $F_y = be^{ax} \sin(by) - xz$, $F_z = -xy - 3z^2$. Let's evaluate the partial differential pairs [1 Mark for each pair, +1 for final statement]:

1. $\frac{\partial F_x}{\partial y} = -x$ and $\frac{\partial F_y}{\partial z} = -x$. Equal, hence conservative.
2. $\frac{\partial F_x}{\partial z} = -y$ and $\frac{\partial F_z}{\partial x} = -y$. Equal, hence conservative.
3. $\frac{\partial F_y}{\partial x} = abe^{ax} \sin(by) - z$ and $\frac{\partial F_z}{\partial y} = abe^{ax} \sin(by) - z$. Equal, hence conservative.

Thus, $\vec{\nabla} \times \vec{F} = \vec{0}$ everywhere, and the field is conservative.

Part b [5 Marks]:

$$dU = (ae^{ax} \cos(by) + yz)dx + (-be^{ax} \sin(by) + xz)dy + (xy + 3z^2)dz \quad [1 \text{ Mark}]$$

Group the terms [2 Marks]:

$$dU = [ae^{ax} \cos(by)dx - be^{ax} \sin(by)dy] + [yzdx + xzdy + xydz] + 3z^2 dz$$

Recognize perfect differentials [1 Mark]:

$$dU = d(e^{ax} \cos(by)) + d(xyz) + d(z^3)$$

Integrate with reference $U(0,0,0) = 0$:

$$U(x, y, z) = -e^{ax} \cos(by) - xyz - z^3 + C$$

Since $U(0, 0, 0) = 0 \implies -1 - 0 - 0 + C = 0 \implies C = 1$.

$$U(x, y, z) = 1 - e^{ax} \cos(by) - xyz - z^3 \quad [1 \text{ Mark}]$$

Part c [3 Marks]: Since the field is conservative, path does not matter [1 Mark].

$$W_{A \rightarrow B} = U(A) - U(B) = U(0, 0, 1) - U(\pi/a, \pi/b, 2) \quad [1 \text{ Mark}]$$

- $U(0, 0, 1) = 1 - e^0 \cos(0) - 0 - 1 = 1 - 1 - 1 = -1$.
- $U(\pi/a, \pi/b, 2) = 1 - e^\pi \cos(\pi) - (\pi/a)(\pi/b)(2) - 8 = 1 + e^\pi - \frac{2\pi^2}{ab} - 8 = e^\pi - \frac{2\pi^2}{ab} - 7$.

$$W_{A \rightarrow B} = -1 - \left(e^\pi - \frac{2\pi^2}{ab} - 7 \right) = 6 - e^\pi + \frac{2\pi^2}{ab} \quad [1 \text{ Mark}]$$

Solution to Problem 3:

Part a [4 Marks]: Extract components: $F_{0,x} = ky$ and $F_{0,y} = kx$. The 2D curl is:

$$\text{Curl}_z = \frac{\partial F_{0,y}}{\partial x} - \frac{\partial F_{0,x}}{\partial y} = k - k = 0 \quad [2 \text{ Marks}]$$

The potential differential is $dU_0 = -F_{0,x}dx - F_{0,y}dy = -kydx - kxdy = -kd(xy)$ [1 Mark]. Integrating with reference $U_0(0,0) = 0$ yields:

$$U_0(x,y) = -kxy \quad [1 \text{ Mark}]$$

Part b [4 Marks]: On the circular track of radius R , the coordinates are $x = R \sin \theta$ and $y = R \cos \theta$ [1 Mark]. The horizontal force is $\vec{F}_0 = kR \cos \theta \hat{i} + kR \sin \theta \hat{j}$. The outward radial unit vector is $\hat{r} = \sin \theta \hat{i} + \cos \theta \hat{j}$. The outward radial component of $\vec{F}_0$ is:

$$F_{0,r} = \vec{F}_0 \cdot \hat{r} = (kR \cos \theta) \sin \theta + (kR \sin \theta) \cos \theta = 2kR \sin \theta \cos \theta \quad [1 \text{ Mark}]$$

Gravity $\vec{F}_g = -mg\hat{j}$ has an inward radial component:

$$F_{g,\text{inward}} = -\vec{F}_g \cdot \hat{r} = mg \cos \theta$$

The inward normal force N plus the gravity inward component minus the outward $\vec{F}_0$ component provides the centripetal force:

$$N + mg \cos \theta - 2kR \sin \theta \cos \theta = \frac{mv^2}{R} \quad [1 \text{ Mark}]$$

Solving for $N(\theta)$ yields:

$$N(\theta) = \frac{mv^2}{R} - mg \cos \theta + 2kR \sin \theta \cos \theta \quad [1 \text{ Mark}]$$

Part c [5 Marks]: Total potential energy is $U(x,y) = mgy + U_0(x,y) = mgy - kxy$ [1 Mark].

- At the top $A(0,R)$, $x = 0, y = R \implies U(A) = mgR - k(0)(R) = mgR$ [1 Mark].
- At the bottom $B(R,0)$, $x = R, y = 0 \implies U(B) = mg(0) - k(R)(0) = 0$ [1 Mark].

Applying conservation of total mechanical energy ($E_A = E_B$) from rest:

$$K_A + U(A) = K_B + U(B) \implies 0 + mgR = \frac{1}{2}mv^2 + 0 \quad [1 \text{ Mark}]$$

$$v = \sqrt{2gR} \quad [1 \text{ Mark}]$$

Solution to Problem 4:

Part a [4 Marks]: Extract $F_x = -y(x^2 + y^2)^{-2}$ and $F_y = x(x^2 + y^2)^{-2}$. Differentiate with respect to x and y [1 Mark for each derivative]:

$$\frac{\partial F_y}{\partial x} = (x^2 + y^2)^{-2} - 2x(2x)(x^2 + y^2)^{-3} = \frac{x^2 + y^2 - 4x^2}{(x^2 + y^2)^3} = \frac{y^2 - 3x^2}{(x^2 + y^2)^3}$$

$$\frac{\partial F_x}{\partial y} = -(x^2 + y^2)^{-2} - (-y)(2)(2y)(x^2 + y^2)^{-3} = \frac{-(x^2 + y^2) + 4y^2}{(x^2 + y^2)^3} = \frac{3y^2 - x^2}{(x^2 + y^2)^3}$$

Subtracting to find Curl [1 Mark]:

$$\text{Curl}_z = \frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} = \frac{y^2 - 3x^2 - 3y^2 + x^2}{(x^2 + y^2)^3} = \frac{-2(x^2 + y^2)}{(x^2 + y^2)^3} = \frac{-2}{(x^2 + y^2)^2}$$

Since $\text{Curl}_z \neq 0$ for any finite coordinate, the field is non-conservative [1 Mark].

Part b [5 Marks]: Since the loop is centered at $(2R, 0)$ and has radius R , it lies entirely in the region $x > 0$. The curl is continuous and well-defined everywhere inside the boundary [1 Mark]. By Stokes' Theorem [1 Mark]:

$$W = \iint \text{Curl}_z dA = \iint_C \frac{-2}{(x^2 + y^2)^2} dA \quad [3 \text{ Marks}]$$

Since Curl is strictly negative, the double integral is non-zero and negative, demonstrating path dependence.

Part c [6 Marks]: For a loop of radius R centered at $(0, 0)$, we parameterize $x = R \cos \theta$, $y = R \sin \theta$ [1 Mark]. The denominator is constant: $x^2 + y^2 = R^2$ [1 Mark]. The work integral is [1 Mark]:

$$W = \oint \frac{-ydx + xdy}{(x^2 + y^2)^2} = \oint \frac{-R \sin \theta (-R \sin \theta d\theta) + R \cos \theta (R \cos \theta d\theta)}{(R^2)^2}$$

$$W = \int_0^{2\pi} \frac{R^2(\sin^2 \theta + \cos^2 \theta)}{R^4} d\theta \quad [2 \text{ Marks}]$$

$$W = \int_0^{2\pi} \frac{1}{R^2} d\theta = \frac{2\pi}{R^2} \quad [1 \text{ Mark}]$$